Math Thermo
class # 24
4/23/2026

## Recapitulation : Two-Phase Incompressible Flow

We have assumed that

1) $r = 2$ (binary mixture),
2) $\Psi_i \equiv 0$, $i = 1, 2$ (no external potential),
3) $\sigma_k \equiv 0$, $k = 1, \dots, K$ (no reactions),
4) $\rho \equiv \rho_0 \equiv$ constant (incompressible).

We know that,

(24.1)
$$\nabla \cdot \vec{v} = 0,$$

(24.2)
$$\rho_0 \frac{\partial \phi_i}{\partial t} + \rho_0 \nabla \cdot (\phi_i \vec{v}) = -\nabla \cdot \vec{J}_i^D, \quad (i = 1, 2)$$

(24.3)
$$\rho_0 \frac{\partial \vec{v}}{\partial t} + \rho_0 \nabla \cdot (\vec{v} \otimes \vec{v}) + \nabla P = -\nabla \cdot \Pi,$$

(24.4)
$$\frac{\partial \hat{u}}{\partial t} + \nabla \cdot (\hat{u} \vec{v}) = -\nabla \cdot \vec{J}_U^D - \Pi : \nabla \vec{v},$$

Our goal is to find all terms is the equation

$$\frac{\partial \hat{s}}{\partial t} + \nabla \cdot (\hat{s} \vec{v}) = -\nabla \cdot \vec{J}_S + \sigma_S.$$

specifically, $J_S^D$ and $\sigma_S$. Here $\hat{s}$ is the entropy density and $\hat{u}$ is the internal energy density.

Under the assumptions that

$$\rho u =: \hat{u} = \tilde{u}(T) + \lambda_u \delta(\phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2),$$

$$\rho s =: \hat{s} = \tilde{s}(T) + \lambda_s \delta(\phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2),$$

where $\lambda_u$ and $\lambda_s$ are constants we have seen that, if $\vec{v}$ is the barycentric velocity,

(24.5)
$$\begin{aligned}\frac{\partial \hat{u}}{\partial t} + \nabla\cdot(\hat{u}\vec{v}) &= \frac{\partial \tilde{u}}{\partial T}\left(\frac{\partial T}{\partial t} + \nabla\cdot(T\vec{v})\right) \\ &+ \lambda_u \omega \left(\frac{\partial \phi}{\partial t} + \nabla\cdot(\phi\vec{v})\right) \\ &+ \lambda_u \varepsilon \nabla\cdot\left(\nabla\phi \frac{\partial \phi}{\partial t}\right) \\ &+ \lambda_u \varepsilon \left[\nabla\cdot(\nabla\phi \otimes \nabla\phi)\right]\cdot\vec{v},\end{aligned}$$

(24.6)
$$\begin{aligned}\frac{\partial \hat{s}}{\partial t} + \nabla\cdot(\hat{s}\vec{v}) &= \frac{\partial \tilde{s}}{\partial T}\left(\frac{\partial T}{\partial t} + \nabla\cdot(T\vec{v})\right) \\ &+ \lambda_s \omega \left(\frac{\partial \phi}{\partial t} + \nabla\cdot(\phi\vec{v})\right) \\ &+ \lambda_s \varepsilon \nabla\cdot\left(\nabla\phi \frac{\partial \phi}{\partial t}\right) \\ &+ \lambda_s \varepsilon \left[\nabla\cdot(\nabla\phi \otimes \nabla\phi)\right]\cdot\vec{v},\end{aligned}$$

where

$$\phi := \phi_2,$$

$$\omega := \frac{1}{\varepsilon}\tilde{W}'(\phi) - \varepsilon\Delta\phi, \qquad \tilde{W}(\phi) := W(1-\phi, \phi).$$

Combining (24.4) and (24.5) we obtained the following heat equation:

(24.7)
$$\frac{\partial \tilde{u}}{\partial T}\left(\frac{\partial T}{\partial t} + \nabla\cdot(T\vec{v})\right) = -\nabla\cdot\vec{J}_U^D - \Pi : \nabla\vec{v}$$
$$-\lambda_u\,\omega\left(\frac{\partial\phi}{\partial t} + \nabla\cdot(\phi\vec{v})\right)$$
$$-\lambda_u\varepsilon\,\nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right)$$
$$-\lambda_u\varepsilon\left[\nabla\cdot(\nabla\phi\otimes\nabla\phi)\right]\cdot\vec{v}.$$

Proposition (24.1): Suppose that thermal compatibility holds, ie,

(24.8)
$$\frac{\partial \tilde{s}}{\partial T} = \frac{1}{T}\frac{\partial \tilde{u}}{\partial T}$$

and assume that

(24.9) $\rho u =: \hat{u}(T, \phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2) = \tilde{u}(T) + \lambda_u\,\delta(\phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2),$

(24.10) $\rho f =: \hat{f}(T, \phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2) = \tilde{f}(T) + \lambda_f\,\delta(\phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2),$

(24.11) $\rho s =: \hat{s}(T, \phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2) = \tilde{s}(T) + \lambda_s\,\delta(\phi_1, \nabla\phi_1, \phi_2, \nabla\phi_2),$

where $\lambda_u$ and $\lambda_s$ are constants and, further,

$$\tilde{f} = \tilde{u} - T\tilde{s} \quad \text{and} \quad \lambda_f = \lambda_u - T\lambda_s.$$

Then,

(24.12)
$$\frac{\partial \hat{s}}{\partial t} + \nabla\cdot(\hat{s}\vec{v}) = -\nabla\cdot\vec{J}_S^D + \sigma_S,$$

where

(24.13)
$$\vec{J}_s^D = \frac{1}{T}J_U^D - \frac{\lambda_f}{\rho_0}\frac{\omega}{T}\vec{J}_2^D + \lambda_f\varepsilon\frac{1}{T}\nabla\phi\frac{\partial\phi}{\partial t} + \lambda_f\varepsilon\frac{1}{T}(\nabla\phi\otimes\nabla\phi)\vec{v},$$

(24.14)
$$\sigma_s = -\frac{1}{T}\nabla\vec{v} : \left(\Pi - \lambda_f\varepsilon\nabla\phi\otimes\nabla\phi\right) - \frac{\lambda_f}{\rho_0}\nabla\left(\frac{\omega}{T}\right)\cdot\vec{J}_2^D + \nabla\left(\frac{1}{T}\right)\cdot\left\{J_U^D + \lambda_f\varepsilon\nabla\phi\frac{\partial\phi}{\partial t} + \lambda_f\varepsilon(\nabla\phi\otimes\nabla\phi)\vec{v}\right\}.$$

<u>Proof</u>: As indicated above

$$\begin{aligned}\frac{\partial\hat{s}}{\partial t} + \nabla\cdot(\hat{s}\vec{v}) = {}& \frac{\partial\tilde{s}}{\partial T}\left(\frac{\partial T}{\partial t} + \nabla\cdot(T\vec{v})\right) \\ &+ \lambda_s\,\omega\left(\frac{\partial\phi}{\partial t} + \nabla\cdot(\phi\vec{v})\right) \\ &+ \lambda_s\varepsilon\nabla\cdot\left(\nabla\phi\frac{\partial\phi}{\partial t}\right) \\ &+ \lambda_s\varepsilon\left[\nabla\cdot(\nabla\phi\otimes\nabla\phi)\right]\cdot\vec{v},\end{aligned}$$

where

$$\omega := \frac{1}{\varepsilon}\tilde{W}'(\phi) - \varepsilon\Delta\phi, \qquad \tilde{W}(\phi) := W(1-\phi,\phi)$$

and

$$\phi := \phi_2 \qquad (\phi_1 = 1-\phi).$$

Using thermal compatibility

$$\begin{aligned}\frac{\partial\hat{s}}{\partial t} + \nabla\cdot(\hat{s}\vec{v}) = {}& \frac{1}{T}\frac{\partial\tilde{u}}{\partial T}\left(\frac{\partial T}{\partial t} + \nabla\cdot(T\vec{v})\right) \\ &+ \lambda_s\,\omega\left(\frac{\partial\phi}{\partial t} + \nabla\cdot(\phi\vec{v})\right) \\ &+ \lambda_s\varepsilon\nabla\cdot\left(\nabla\phi\frac{\partial\phi}{\partial t}\right)\end{aligned}$$

$$+\lambda_s \varepsilon \left[\nabla\cdot\left(\nabla\phi\otimes\nabla\phi\right)\right]\cdot\vec{v},$$

or, equivalently,

(24.15)
$$\begin{aligned} T\left(\frac{\partial\hat{s}}{\partial t}+\nabla\cdot\left(\hat{s}\,\vec{v}\right)\right) &= \frac{\partial\tilde{u}}{\partial T}\left(\frac{\partial T}{\partial t}+\nabla\cdot\left(T\vec{v}\right)\right) \\ &\quad + T\lambda_s\,\omega\left(\frac{\partial\phi}{\partial t}+\nabla\cdot\left(\phi\,\vec{v}\right)\right) \\ &\quad + T\lambda_s\varepsilon\,\nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right) \\ &\quad + T\lambda_s\varepsilon\left[\nabla\cdot\left(\nabla\phi\otimes\nabla\phi\right)\right]\cdot\vec{v}. \end{aligned}$$

Combining

$$\frac{\partial\hat{u}}{\partial t}+\nabla\cdot\left(\hat{u}\,\vec{v}\right) = -\nabla\cdot\vec{J}_U^{\,0} - \Pi:\nabla\vec{v}$$

and

$$\begin{aligned} \frac{\partial\hat{u}}{\partial t}+\nabla\cdot\left(\hat{u}\,\vec{v}\right) &= \frac{\partial\tilde{u}}{\partial T}\left(\frac{\partial T}{\partial t}+\nabla\cdot\left(T\vec{v}\right)\right) \\ &\quad + \lambda_u\,\omega\left(\frac{\partial\phi}{\partial t}+\nabla\cdot\left(\phi\,\vec{v}\right)\right) \\ &\quad + \lambda_u\varepsilon\,\nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right) \\ &\quad + \lambda_u\varepsilon\left[\nabla\cdot\left(\nabla\phi\otimes\nabla\phi\right)\right]\cdot\vec{v} \end{aligned}$$

we recall the following form of the heat equation:

$$\frac{\partial \tilde{u}}{\partial T}\left(\frac{\partial T}{\partial t}+\nabla\cdot(T\vec{v})\right) = -\nabla\cdot\vec{J}_U^D - \Pi:\nabla\vec{v}$$
$$-\lambda_u\,\omega\left(\frac{\partial\phi}{\partial t}+\nabla\cdot(\phi\vec{v})\right)$$
(24.16)
$$-\lambda_u\varepsilon\nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right)$$
$$-\lambda_u\varepsilon\left[\nabla\cdot(\nabla\phi\otimes\nabla\phi)\right]\cdot\vec{v}.$$

Substituting (24.16) into (24.15) we have

$$T\left(\frac{\partial\hat{s}}{\partial t}+\nabla\cdot(\hat{s}\vec{v})\right) = -\nabla\cdot\vec{J}_U^D - \Pi:\nabla\vec{v}$$
$$+(T\lambda_s-\lambda_u)\,\omega\left(\frac{\partial\phi}{\partial t}+\nabla\cdot(\phi\vec{v})\right)$$
$$+(T\lambda_s-\lambda_u)\,\varepsilon\nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right)$$
$$+(T\lambda_s-\lambda_u)\,\varepsilon\left[\nabla\cdot(\nabla\phi\otimes\nabla\phi)\right]\cdot\vec{v}$$
$$= -\nabla\cdot\vec{J}_U^D - \Pi:\nabla\vec{v}$$
$$-\lambda_f\,\omega\left(\frac{\partial\phi}{\partial t}+\nabla\cdot(\phi\vec{v})\right)$$
$$-\lambda_f\varepsilon\nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right)$$
$$-\lambda_f\,\varepsilon\left[\nabla\cdot(\nabla\phi\otimes\nabla\phi)\right]\cdot\vec{v}.$$

Using the component mass conservation equation

$$\frac{\partial\phi}{\partial t}+\nabla\cdot(\phi\vec{v}) = -\frac{\nabla\cdot\vec{J}_2^D}{\rho_0},$$

we have

$$T\left(\frac{\partial \hat{s}}{\partial t}+\nabla\cdot\left(\hat{s}\vec{v}\right)\right) = -\nabla\cdot\vec{J}_U^D - \Pi : \nabla\vec{v} + \lambda_f\,\omega\,\frac{\nabla\cdot\vec{J}_2^D}{\rho_0} - \lambda_f\,\varepsilon\,\nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right) - \lambda_f\,\varepsilon\left[\nabla\cdot\left(\nabla\phi\otimes\nabla\phi\right)\right]\cdot\vec{v}. \tag{24.17}$$

Equivalently,

$$\frac{\partial \hat{s}}{\partial t}+\nabla\cdot\left(\hat{s}\vec{v}\right) = \frac{-\nabla\cdot\vec{J}_U^D}{T} \;\frac{-\Pi : \nabla\vec{v}}{T} + \frac{\lambda_f\,\omega}{\rho_0 T}\,\nabla\cdot\vec{J}_2^D - \frac{\lambda_f\,\varepsilon}{T}\nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right) - \frac{\lambda_f\,\varepsilon}{T}\left[\nabla\cdot\left(\nabla\phi\otimes\nabla\phi\right)\right]\cdot\vec{v}. \tag{24.18}$$

Now, we use some identities:

$$\frac{\omega}{T}\nabla\cdot\vec{J}_2^D = \nabla\cdot\left(\frac{\omega}{T}\vec{J}_2^D\right) - \nabla\left(\frac{\omega}{T}\right)\cdot\vec{J}_2^D,$$

$$\frac{1}{T}\nabla\cdot\vec{J}_U^D = \nabla\cdot\left(\frac{1}{T}\vec{J}_U^D\right) - \nabla\left(\frac{1}{T}\right)\cdot\vec{J}_U^D,$$

$$\frac{1}{T}\nabla\cdot\left(\nabla\phi\frac{\partial\phi}{\partial t}\right) = \nabla\cdot\left(\frac{1}{T}\nabla\phi\frac{\partial\phi}{\partial t}\right) - \nabla\left(\frac{1}{T}\right)\cdot\nabla\phi\,\frac{\partial\phi}{\partial t},$$

$$\left[\nabla\cdot\left(\nabla\phi\otimes\nabla\phi\right)\right]\cdot\vec{v} = \nabla\cdot\left[\left(\nabla\phi\otimes\nabla\phi\right)\vec{v}\right] - \left(\nabla\phi\otimes\nabla\phi\right):\nabla\vec{v}.$$

Inserting into (24.18), we find

(24.19)
$$\frac{\partial \hat{s}}{\partial t} + \nabla\cdot(\hat{s}\vec{v}) = -\nabla\cdot\left(\frac{1}{T}\vec{J}_u^D\right) + \nabla\left(\frac{1}{T}\right)\cdot\vec{J}_u^D + \frac{\lambda_f}{\rho_0}\nabla\cdot\left(\frac{\omega}{T}\vec{J}_2^D\right) - \frac{\lambda_f}{\rho_0}\nabla\left(\frac{\omega}{T}\right)\cdot\vec{J}_2^D - \lambda_f\varepsilon\nabla\cdot\left(\frac{1}{T}\nabla\phi\frac{\partial\phi}{\partial t}\right) + \lambda_f\varepsilon\nabla\left(\frac{1}{T}\right)\cdot\nabla\phi\frac{\partial\phi}{\partial t} - \frac{\lambda_f\varepsilon}{T}\nabla\cdot\left[(\nabla\phi\otimes\nabla\phi)\vec{v}\right] + \frac{\lambda_f\varepsilon}{T}(\nabla\phi\otimes\nabla\phi):\nabla\vec{v} - \frac{\Pi:\nabla\vec{v}}{T}.$$

One more identity should do it:

$$\frac{1}{T}\nabla\cdot\left[(\nabla\phi\otimes\nabla\phi)\vec{v}\right] = \nabla\cdot\left[\frac{1}{T}(\nabla\phi\otimes\nabla\phi)\vec{v}\right] - \nabla\left(\frac{1}{T}\right)\cdot\left[(\nabla\phi\otimes\nabla\phi)\vec{v}\right].$$

Then (24.19) becomes

(24.20)
$$\frac{\partial \hat{s}}{\partial t} + \nabla\cdot(\hat{s}\vec{v}) = -\nabla\cdot\left(\frac{1}{T}\vec{J}_u^D\right) + \nabla\left(\frac{1}{T}\right)\cdot\vec{J}_u^D + \frac{\lambda_f}{\rho_0}\nabla\cdot\left(\frac{\omega}{T}\vec{J}_2^D\right) - \frac{\lambda_f}{\rho_0}\nabla\left(\frac{\omega}{T}\right)\cdot\vec{J}_2^D - \lambda_f\varepsilon\nabla\cdot\left(\frac{1}{T}\nabla\phi\frac{\partial\phi}{\partial t}\right) + \lambda_f\varepsilon\nabla\left(\frac{1}{T}\right)\cdot\nabla\phi\frac{\partial\phi}{\partial t} + \frac{\lambda_f\varepsilon}{T}(\nabla\phi\otimes\nabla\phi):\nabla\vec{v} - \frac{\Pi:\nabla\vec{v}}{T} - \lambda_f\varepsilon\nabla\cdot\left[\frac{1}{T}(\nabla\phi\otimes\nabla\phi)\vec{v}\right] + \lambda_f\varepsilon\nabla\left(\frac{1}{T}\right)\cdot\left[(\nabla\phi\otimes\nabla\phi)\vec{v}\right].$$

Comparing (24.20) to (24.12), we find

$$\vec{J}_s^D = \frac{1}{T}J_v^D - \frac{\lambda_f}{\rho_0}\frac{\omega}{T}\vec{J}_2^D + \lambda_f \varepsilon \frac{1}{T}\nabla\phi\frac{\partial\phi}{\partial t} + \lambda_f \varepsilon \frac{1}{T}(\nabla\phi \otimes \nabla\phi)\vec{v}$$

$$\sigma_s = -\frac{1}{T}\nabla\vec{v} : \left(\Pi - \lambda_f\varepsilon\nabla\phi\otimes\nabla\phi\right) - \frac{\lambda_f}{\rho_0}\nabla\left(\frac{\omega}{T}\right)\cdot\vec{J}_2^D + \nabla\left(\frac{1}{T}\right)\cdot\left\{J_v^D + \lambda_f\varepsilon\nabla\phi\frac{\partial\phi}{\partial t} + \lambda_f\varepsilon(\nabla\phi\otimes\nabla\phi)\vec{v}\right\}$$

Thus (24.13) and (24.14) are confirmed. ///

Other forms are possible. Consider the following identity:

$$\nabla\left(\frac{\omega}{T}\right)\cdot\vec{J}_2^D = \frac{1}{T}\nabla\omega\cdot\vec{J}_2^D + \nabla\left(\frac{1}{T}\right)\cdot\omega\vec{J}_2^D.$$

Thus, we have the following:

<u>Corollary</u> (24.2): Under the same conditions as in the last proposition

$$\frac{\partial\hat{s}}{\partial t} + \nabla\cdot(\hat{s}\vec{v}) = -\nabla\cdot\vec{J}_s^D + \sigma_s,$$

where

(24.21)
$$\vec{J}_s^D = \frac{1}{T}J_v^D - \frac{\lambda_f}{\rho_0}\frac{\omega}{T}\vec{J}_2^D + \lambda_f\varepsilon\frac{1}{T}\nabla\phi\frac{\partial\phi}{\partial t} + \lambda_f\varepsilon\frac{1}{T}(\nabla\phi\otimes\nabla\phi)\vec{v}$$

and

(24.22) $$\sigma_s = -\frac{1}{T}\nabla\vec{v} : \left(\Pi - \lambda_f \varepsilon \nabla\phi \otimes \nabla\phi\right) - \frac{\lambda_f}{\rho_0 T}\nabla\omega \cdot \vec{J}_2^D + \nabla\left(\frac{1}{T}\right)\cdot\left\{ J_v^D + \lambda_f \varepsilon \nabla\phi \frac{\partial\phi}{\partial t} + \lambda_f \varepsilon (\nabla\phi \otimes \nabla\phi)\vec{v} - \frac{\lambda_f}{\rho_0}\omega \vec{J}_2^D \right\}.$$

## Constitutive Relations

Now, finally, we can choose $\Pi$, $\vec{J}_2^D$, and $J_v^D$ in order to ensure that

$$\sigma_s \geq 0,$$

consistent with the second law of thermodynamics.

There are many ways to do this, as we shall see. For now, we take the simplest path forward.

**Theorem** (24.3) Under the assumptions of Proposition (24.1)

$$\sigma_s \geq 0$$

is guaranteed, provided

(24.23) $$\Pi = -\mu \tau + \lambda_f \varepsilon \nabla\phi \otimes \nabla\phi$$

where $\mu > 0$ and $\tau$ is the (deviatoric) shear rate tensor

$$\tau := \frac{1}{2}\left(\nabla\vec{v} + (\nabla\vec{v})^T\right) - \frac{1}{3}\nabla\cdot\vec{v}\, I_3$$

$$= \frac{1}{2}\left(\nabla\vec{v} + (\nabla\vec{v})^T\right),$$

(24.24) $$\vec{J}_2^D = - M_2 \nabla \omega,$$

where $M_2 > 0$ is a constant; and

(24.25) $$J_U^D = M_T T^2 \nabla\left(\frac{1}{T}\right) - \lambda_f \varepsilon \nabla\phi \frac{\partial \phi}{\partial t} - \lambda_f \varepsilon (\nabla\phi \otimes \nabla\phi)\vec{v} - \frac{\lambda_f M_2}{\rho_0} \omega \nabla \omega,$$

where $M_T > 0$ is a constant. In this case

(24.26) $$\sigma_S = \frac{\mu}{T} \nabla\vec{v} : \tau + \frac{\lambda_f}{\rho_0 T} M_2 |\nabla\omega|^2 + M_T T^2 \left|\nabla\left(\frac{1}{T}\right)\right|^2 \geq 0.$$

Proof: Substitute (24.23) – (24.25) into (24.22) to arrive at (24.26). Note that

$$\nabla\vec{v} : \tau \geq 0$$

appealing to Proposition (22.2). ///

Remark (24.4): Note that the positivity of the entropy production term relies on the fact the temperature function remains positive. This is a non-trivial assumption.

# The Complete Model

Now that the constitution relations have been chosen we can write the models in detail.

Let us assume that

$$\tilde{u} = \rho_0 C_n T,$$
$$\tilde{f} = \rho_0 C_n T - \rho_0 C_n T \ln\left(\frac{T}{T_0}\right),$$
$$\tilde{s} = \rho_0 C_n \ln\left(\frac{T}{T_0}\right).$$

As we have observed, thermal compatibility holds for this model

$$\frac{\partial \tilde{s}}{\partial T} = \frac{1}{T} \frac{\partial \tilde{u}}{\partial T}$$

and, clearly,

$$\tilde{f} = \tilde{u} - T\tilde{s}.$$

The important fact for the final form of our equations is that

$$\frac{\partial \tilde{u}}{\partial T} = \rho_0 C_n > 0$$

is a positive constant. $C_n$ is known as the <u>heat capacity</u>.

The dynamical equations are, in all the details, (24.27) $$\nabla\cdot\vec{v} = 0,$$ (Continuity)

(24.28) $$\frac{\partial\phi}{\partial t} + \nabla\cdot(\phi\,\vec{v}) = \frac{M}{\rho_0}\Delta\omega,$$ (Cahn-Hilliard)

(24.29) $$\omega = \frac{1}{\varepsilon}\widetilde{W}'(\phi) - \varepsilon\Delta\phi,$$

(24.30) $$\rho_0\left(\frac{\partial\vec{v}}{\partial t} + \nabla\cdot(\vec{v}\otimes\vec{v})\right) + \nabla P = \nabla\cdot(\mu\tau) - \lambda_f\varepsilon\nabla\cdot(\nabla\phi\otimes\nabla\phi)$$

(Navier-Stokes)

(24.31) $$\tau = \frac{1}{2}\left(\nabla\vec{v} + (\nabla\vec{v})^T\right),$$

(24.32) $$\lambda_f = \lambda_u - T\lambda_s,$$

(24.33) (heat (Fourier) equation)

$$\begin{aligned}
\rho_0 C_h\left(\frac{\partial T}{\partial t} + \nabla\cdot(T\vec{v})\right) = {} & M_T\,\Delta T + \lambda_f\varepsilon\,\nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right) \\
& + \lambda_f\varepsilon\,\nabla\cdot\left[(\nabla\phi\otimes\nabla\phi)\,\vec{v}\right] \\
& + \frac{\lambda_f M_2}{\rho_0}\nabla\cdot(\omega\nabla\omega) \\
& - \nabla\vec{v} : \left[\mu\tau - \lambda_f\varepsilon\,\nabla\phi\otimes\nabla\phi\right] \\
& - \lambda_u\,\omega\left(\frac{\partial\phi}{\partial t} + \nabla\cdot(\phi\,\vec{v})\right) \\
& - \lambda_u\varepsilon\,\nabla\cdot\left(\nabla\phi\,\frac{\partial\phi}{\partial t}\right) \\
& - \lambda_u\varepsilon\left[\nabla\cdot(\nabla\phi\otimes\nabla\phi)\right]\cdot\vec{v}.
\end{aligned}$$

This is called the (or a)

<u>Cahn-Hilliard-Navier-Stokes-Fourier</u>

model.

# Onsager – Casimir Relations

**Theorem** (24.5): Suppose that

$$\sigma_s = \sum_{i=1}^{n} J_i^\alpha X_i^\alpha + \sum_{i=1}^{m} J_i^\beta X_i^\beta,$$

where $J_i^\alpha, J_i^\beta$ are fluxes and $X_i^\alpha, X_i^\beta$ are even ($\alpha$) and odd ($\beta$) thermodynamic forces. Assume that

$$J_i^\alpha = \sum_{j=1}^{n} L_{i,j}^{\alpha,\alpha} X_j^\alpha + \sum_{j=1}^{m} L_{i,j}^{\alpha,\beta} X_j^\beta, \qquad 1 \le i \le n,$$

$$J_i^\beta = \sum_{j=1}^{n} L_{i,j}^{\beta,\alpha} X_j^\alpha + \sum_{j=1}^{m} L_{i,j}^{\beta,\beta} X_j^\beta, \qquad 1 \le i \le m,$$

where

(24.34) $$L_{i,j}^{\alpha,\alpha} = L_{j,i}^{\alpha,\alpha}, \qquad 1 \le i,j \le n,$$

(24.35) $$L_{i,j}^{\beta,\beta} = L_{j,i}^{\beta,\beta}, \qquad 1 \le i,j \le m,$$

(24.36) $$L_{i,j}^{\alpha,\beta} = -L_{j,i}^{\beta,\alpha}, \qquad 1 \le i \le n, \quad 1 \le j \le m.$$

Set

$$L := \begin{bmatrix} L^{\alpha,\alpha} & L^{\alpha,\beta} \\ L^{\beta,\alpha} & L^{\beta,\beta} \end{bmatrix}, \qquad J := \begin{bmatrix} \vec{J}^\alpha \\ \vec{J}^\beta \end{bmatrix}, \qquad X := \begin{bmatrix} \vec{X}^\alpha \\ \vec{X}^\beta \end{bmatrix}.$$

Then

$$\sigma_s = X^T L_s X = \vec{X}^{\alpha^T} L^{\alpha,\alpha} \vec{X}^\alpha + \vec{X}^{\beta^T} L^{\beta,\beta} \vec{X}^\beta,$$

where

$$L_s = \tfrac{1}{2}\left(L + L^T\right) = \begin{bmatrix} L^{\alpha,\alpha} & 0 \\ 0 & L^{\beta,\beta} \end{bmatrix}.$$

*Proof:* Clearly,

$$\sigma_s = \vec{J}^T \vec{X}$$

and

$$\vec{J} = L \vec{X}.$$

In particular,

$$\vec{J} = L\vec{X} = \begin{bmatrix} L^{\alpha,\alpha}\vec{X}^{\alpha} + L^{\alpha,\beta}\vec{X}^{\beta} \\ L^{\beta,\alpha}\vec{X}^{\alpha} + L^{\beta,\beta}\vec{X}^{\beta} \end{bmatrix}.$$

Thus, since $L^{\alpha,\alpha^T} = L^{\alpha,\alpha}$ and $L^{\beta,\beta^T} = L^{\beta,\beta}$,

$$\vec{J}^T = \left[\vec{X}^{\alpha^T} L^{\alpha,\alpha} + \vec{X}^{\beta^T} L^{\alpha,\beta^T},\ \vec{X}^{\alpha^T} L^{\beta,\alpha^T} + \vec{X}^{\beta^T} L^{\beta,\beta}\right]$$

and

$$\begin{aligned}
\sigma_s &= \vec{J}^T \vec{X} \\
&= \vec{X}^{\alpha^T} L^{\alpha,\alpha} \vec{X}^{\alpha} + \vec{X}^{\beta^T} L^{\alpha,\beta^T} \vec{X}^{\alpha} \\
&\quad + \vec{X}^{\alpha^T} L^{\beta,\alpha^T} \vec{X}^{\beta} + \vec{X}^{\beta^T} L^{\beta,\beta} \vec{X}^{\beta} \\
&= \vec{X}^{\alpha^T} L^{\alpha,\alpha} \vec{X}^{\alpha} + \vec{X}^{\alpha^T} L^{\alpha,\beta} \vec{X}^{\beta} \\
&\quad + \vec{X}^{\alpha^T} L^{\beta,\alpha^T} \vec{X}^{\beta} + \vec{X}^{\beta^T} L^{\beta,\beta} \vec{X} \\
&= \vec{X}^{\alpha^T} L^{\alpha,\alpha} \vec{X}^{\alpha} + \vec{X}^{\alpha^T} L^{\alpha,\beta} \vec{X}^{\beta} \\
&\quad - \vec{X}^{\alpha^T} L^{\alpha,\beta} \vec{X}^{\beta} + \vec{X}^{\beta^T} L^{\beta,\beta} \vec{X} \\
&= \vec{X}^{\alpha^T} L^{\alpha,\alpha} \vec{X}^{\alpha} + \vec{X}^{\beta^T} L^{\beta,\beta} \vec{X},
\end{aligned}$$

since

$$L^{\beta,\alpha^T} = -L^{\alpha,\beta}. \quad ///$$

The relations (24.34) – (24.36) are called <u>Onsager – Casimir</u> relations.

<u>Corollary</u> (24.6): If $L^{\alpha,\alpha}$ and $L^{\beta,\beta}$ are positive semi-definite, then $\sigma_S \geq 0$.